

Software Requirements Specification

for

School Management System

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1. Introduction

Purpose

This Software Requirements Specification (SRS) document outlines the functional and non-functional requirements for the **School Management System (SMS)**, version 1.0. The purpose of this system is to streamline and automate the core administrative and academic processes of a school. It is designed to manage student information, admissions, fee payments, exam schedules, results, staff data, and more in an efficient and centralized manner.

Project Scope

The **School Management System (SMS)** is a comprehensive software solution designed to digitalize and automate the core administrative, academic, and financial operations within a school environment. The primary goal of this system is to simplify day-to-day tasks such as managing student records, admissions, fee transactions, exam scheduling, and result processing, thereby reducing manual workload and enhancing efficiency across departments.

Overall Description

Product Perspective

- ☐ Student Management
- ☐ Fee Management
- ☐ Exam and Result Scheduling
- ☐ Staff Records

Product Features

1. Student Management
2. Fee Management
3. Exam and Result Management
4. Staff Management
5. Search
6. Security and Validation

User Classes and Characteristics

1. Account/Finance Staff
2. Teachers
3. Student

Operating Environment

Hardware Requirements:
Client Machine

Software Requirements:
Microsoft Windows 10/11 (64-bit recommended)

Development Framework:
.NET Framework 4.7.2 or later

Database Management System:
Microsoft SQL Server 2019 or later

IDE
Visual Studio 2019 or later

User Documentation

- ☐ All documentation will be provided in **PDF format** for easy access and printing.
- ☐ Documentation will be included in the software installation folder and can also be delivered via email or USB on deployment.
- ☐ On-screen tooltips will be embedded within the WinForms application itself.

Assumptions and Dependencies

Assumptions

1. Single Institution Use
2. Windows Operating System Availability
3. Trained Users
4. Pre-installed SQL Server

Dependencies

Microsoft SQL Server
Existing Student Data

System Features

- 1. Admin Panel Features**
 - 1.1 Admission Management
 - 1.2 1.2 Fee Management
 - 1.3 1.3 Teacher Management
 - 1.4 1.4 Search Functionality
 - 1.5 1.5 Student & Teacher Details
- 2. Teacher Panel Features**
 - 2.1 Class Schedule Management
 - 2.2 Student Admission
 - 2.3 Attendance Management
 - 2.4 Registration
 - 2.5 Result Management
 - 2.6 Document Upload
- 3. Student Panel Features**
 - 3.1 Class Schedule Viewing
 - 3.2 Exam Schedule
 - 3.3 View Documents
 - 3.4 Result Viewing

External Interface Requirements

User Interfaces

- **UI Components Needed**
- Login Interface
- Admin Dashboard & Submodules
- Teacher Panel Interfaces
- Student View Panels
- Modal popups for data entry/editing
- Confirmation and alert boxes

Hardware Interfaces

☐ **Supported Devices:**

Windows-based desktops/laptops (Windows 10+), printers (USB/Wi-Fi), barcode scanners (optional), network routers/switches.

☐ **Minimum Requirements:**

- CPU: Intel i3+
- RAM: 4 GB (8 GB recommended)
- Storage: 500 MB+
- Display: 1366x768+

☐ **Input/Output:**

Keyboard, mouse, monitor, optional touchscreen, printer (for receipts/reports), USB storage for backups.

☐ **Communication:**

- SQL Server via TCP/IP

Software Interfaces

☐ **OS:** Windows 10 or later

☐ **Database:** MS SQL Server 2019 (centralized data storage)

☐ **Language & Framework:** C# with .NET Framework 4.7+

☐ **Libraries:** System.Data.SqlClient, Report Viewer, optional Crystal Reports

☐ **Data Flow:** User inputs (e.g., admissions, attendance) → stored in DB → outputs (e.g., reports, schedules)

☐ **Communication:** SQL over TCP/IP, secure role-based access

Other Nonfunctional Requirements

Performance Requirements

1.1 Performance Requirements

- The system should load all major modules (e.g., admission, attendance) within 3 seconds.
- Database queries must respond within 2 seconds under normal load.
- The system should support 10+ concurrent users without lag.
- Reports (e.g., results, receipts) should generate within 5 seconds.
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Safety Requirements

- ☐ Regular data backups must be performed to prevent data loss.
- ☐ Unauthorized access to sensitive data (e.g., student records, financial info) must be restricted.
- ☐ Admin actions (e.g., delete/update records) should require confirmation prompts.
- ☐ The system should follow standard data protection policies (e.g., school/institutional IT policies).
- ☐ Error logs must be maintained to detect and resolve potential risks.

Security Requirements

- Role-based access control must be enforced (Admin, Teacher, Student).
- Users must log in with unique credentials; passwords must be encrypted.
- Sensitive data (e.g., fees, results) must be accessible only by authorized users.
- Session timeouts should occur after a period of inactivity.
- The system must comply with institutional data privacy policies.

Software Quality Attributes

- ☐ **Usability:** Easy-to-navigate UI for admins, teachers, and students.
- ☐ **Reliability:** System should run without failure during school hours.
- ☐ **Maintainability:** Code should be modular and easy to update.
- ☐ **Security:** Data access strictly based on user roles and credentials.
- ☐ **Availability:** 99% uptime during academic operations.
- ☐ **Portability:** Should run on any Windows 10+ machine without reconfiguration.
- ☐ **Scalability:** System should handle increased users and records over time.